Agentic Knowledge Graphs for Research Progression

PhD Qualifying Exam Presentation

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Outline

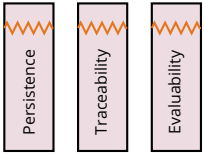
1. Motivation
2. Literature Survey
3. Synthesis
4. Proposal
5. Evaluation
6. Contributions
7. Conclusion

The Structural Problem in AI-Assisted Research

- Research reasoning is **fragmented** and **transient**
- LLM-assisted workflows **amplify volatility**
- **Structural memory** is missing

The Structural Gaps

- **Persistence:** State beyond session boundaries
- **Traceability:** Linkage from Claim $\rightarrow$ Evidence
- **Evaluability:** Empirical measurement of progression



These are infrastructure problems, not model-quality problems.

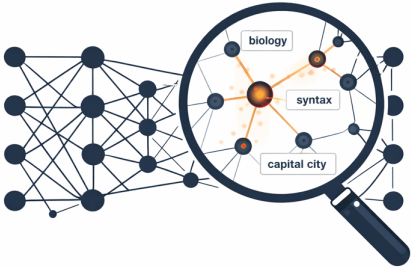
Guiding Question

How can research state be made persistent, traceable, and empirically evaluable in AI-assisted workflows?

Landscape Overview

Mechanistic Interpretability

- Reverse-engineer the black box[a]
- Sparse features → human-interpretable structure
- Tells us *how* a model thinks

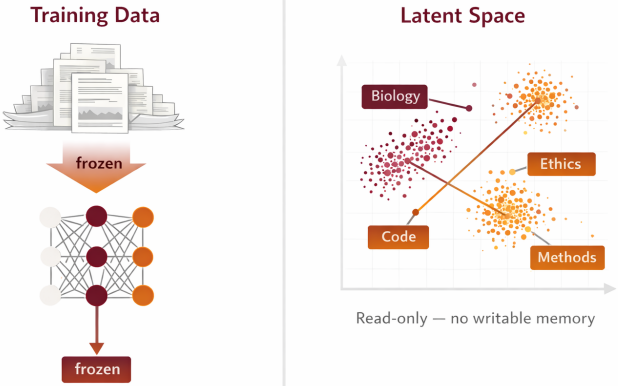


Limitation

[a] Sharkey et al. 2025.

Static weights only — no writable memory for new research state.

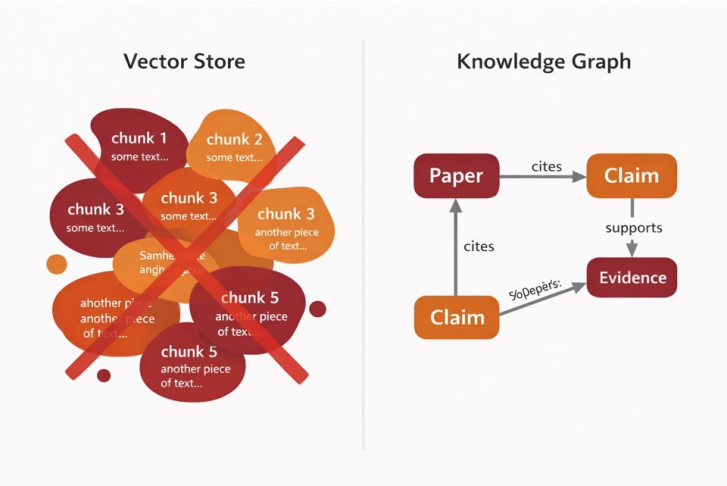
Concept-Based Interpretability



Limitation

Read-only — no mechanism to encode new research state.

RAG & Vector Memory Systems



Limitations[a]

[a] Suryawanshi et al. 2025.

- No canonical memory
- No provenance chain
- No structured reasoning
- No longitudinal state

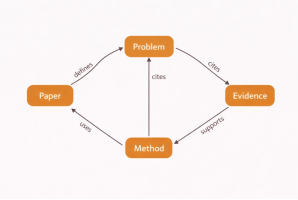
Knowledge Graph Research

Strengths

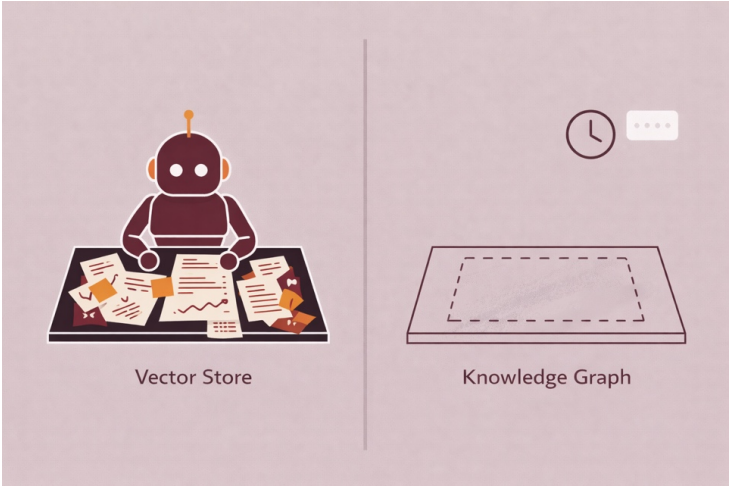
- Ontology-based structure
- Provenance modeling
- Logical relationships

Limitations

- Manual curation
- Static graphs
- No automated maintenance



Agentic Systems



Limitations

- Ephemeral memory
- No structural persistence
- Weak auditability

Empirical Evaluation Standards

- Ralph et al. (2021): Empirical Standards for SE
- Baltes et al. (2025): Guidelines for LLM Studies

Observation

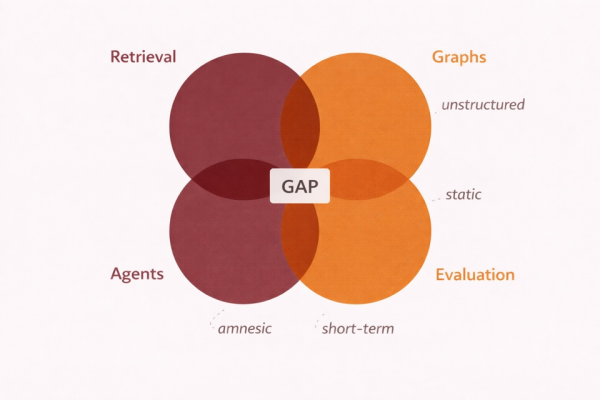
Single-pass or qualitative evaluations dominate.

Need

Longitudinal evaluation of coherent research state over time.

- [x] Standards defined
- [x] Guidelines published
- [ ] Agent validation?
- [ ] Longitudinal metrics?
- [ ] Provenance audit?

Cross-Cutting Gaps in the Literature



- **Retrieval** is unstructured
- **Graphs** are static
- **Agents** are amnesic
- **Evaluation** is short-term

This fragmentation motivates new infrastructure.

Positioning My Work

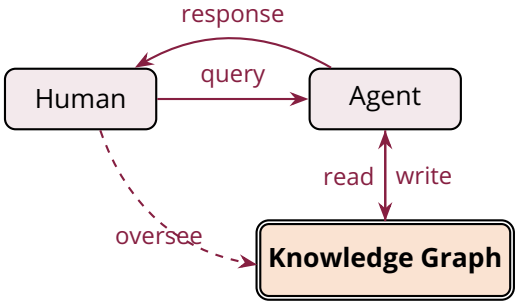
Intersection

Agentic systems + Knowledge graph persistence + Span-level traceability

Literature Refined:

- Stronger empirical evaluation
- SE domain scope
- Longitudinal validation

Proposed Architecture

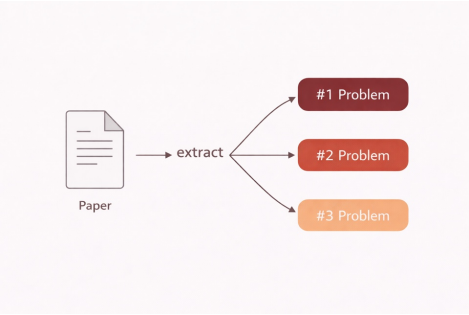


Persistent Research State Loop

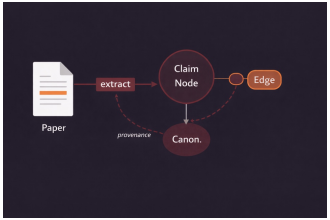
- Human-in-the-loop oversight
- Agent-mediated updates
- KG as **Source of Truth**

Problem Extraction & Ranking

- Papers → problem nodes in KG
- Ranked by evidence density & citation weight
- Rank guides agent focus



Core Innovation



1. **Span-level alignment** — every node traces to a source sentence
2. **Canonicalization Agent** — de-duplicates entities in the graph
3. **Versioned provenance chains** — append-only audit trail

Evaluation Framework

Alignment Canonicalization Traceability Persistence

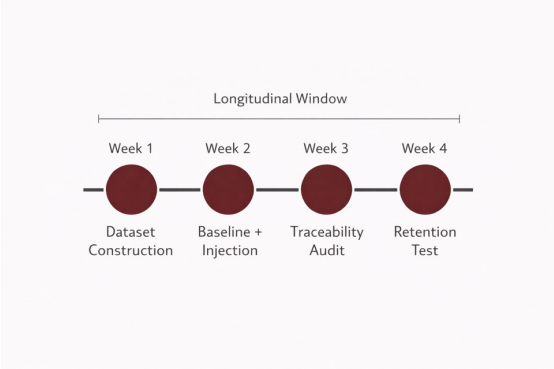
Claim	Metric	Baseline
Span-level alignment	Precision / Recall / F1	Standard RAG
Canonicalization	Cohen's κ	Embedding clustering
Traceability	Audit completeness %	RAG w/ citations
Persistence	Retention accuracy	Chat-only LLM

Key Question

Does the system retain knowledge better than a chat-only LLM across multiple sessions?

Experimental Design Overview

- **Dataset:** Annotated SE spans
- **Injection:** Synthetic duplicates
- **Audit:** Traceability check
- **Task:** Retention measurement



Contributions

Span-level Specification

Curation Protocol

Reference Infrastructure

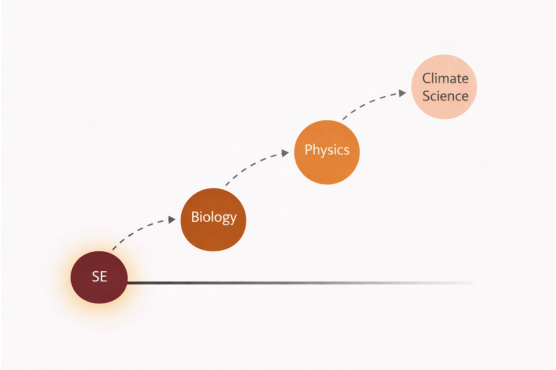
1. **Span-level traceability specification** — foundational vocabulary for the field
2. **Agentic Curation Protocol** — how agents write, update, and validate the KG
3. **Reference infrastructure** — open-source implementation for benchmarking

Near-Term Work

- *Months 1–6:* Build system
- *Month 6–7:* Evaluation study
- *Month 7–8:* Artifact release
- *Year 1:* SE deployment

Long-Term Vision

- Cross-domain expansion (Biology, Physics, Climate Science)
- Persistent AI research infrastructure
- **Cross-domain synthesis:** same problem, different fields



Conclusion

- **From:** Fragmented, ephemeral chat
- **To:** Persistent, traceable state
- **Goal:** Infrastructure for discovery

Infrastructure is the missing link to autonomous scientific discovery.

Questions?

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Backup Slides

Anticipated Questions — Motivation & Literature

Q [S2]: How is Evaluability different from accuracy? Accuracy measures a single output. Evaluability means the system exposes enough structure to measure *longitudinal* correctness — whether knowledge degrades, drifts, or accumulates errors across sessions.

Q [S7]: Have you considered hybrid RAG + KG approaches? Yes — hybrid systems exist (GraphRAG, etc.) but they use the KG only for retrieval routing, not as an auditable research state. Our contribution is provenance-first: every node is anchored to a source span regardless of retrieval method.

Q [S8]: How does your KG differ from existing domain KGs (Wikidata, biomedical)? Domain KGs are manually curated, static, and general-purpose. Ours is agent-maintained, append-only versioned, and scoped to the research workflow — claims link to specific paper spans, not to universal entity definitions.

Anticipated Questions — Literature (cont.)

Q [S9]: How does your system avoid the same amnesia you critique? By externalizing state into the KG. The agent's context window can reset; the graph does not. Every session reads from the graph first, so the agent starts informed, not cold.

Q [S12]: How does this compare to LangChain Memory or MemGPT? Those frameworks provide session-scoped buffers or rolling summaries — unstructured, unverifiable, and not provenance-aware. Our KG is structured, auditable, and verifiable against source spans.

Anticipated Questions — Architecture

Q [S13]: How does human oversight work in practice? The human reviews the graph state via a query interface — seeing which claims were added, what spans support them, and flagging incorrect merges. It is not continuous monitoring; it is checkpoint-based review.

Q [S13b]: How do you define a "research problem" vs. a finding or method? A problem node captures an open challenge stated or implied in the source. Findings are answers; methods are approaches. The extraction agent uses typed predicates to distinguish them in the ontology.

Q [S14]: How do provenance chains work? Each KG write records the source span, the agent, and the prior graph state — like a git commit log. An append-only versioned log; blockchain is a future scaling option.

Q [S14]: Why not just use RAG? RAG retrieves chunks, not structure. It has no canonical memory, no de-duplication, and no provenance. Our system adds all three.

Anticipated Questions — Evaluation

Q [S15]: How is your baseline RAG configured — does it also get citations? Yes — the RAG baseline includes citation-level retrieval (RAG w/ citations) so we are comparing against the strongest plausible alternative, not a strawman.

Q [S16]: Why 4 weeks? Why not longer? Four weeks is the minimum viable window to observe retention patterns; consistent with longitudinal SE field study norms (Ralph, Baltes). The protocol extends naturally to longer periods.

Q [S16]: What's your implementation timeline? Six months to build a working prototype, followed immediately by the 4-week evaluation study. Artifact release and SE deployment in Year 1.

Anticipated Questions — Contributions

Q [S17]: What does the formal Research State definition look like? It is operationalized through the architecture: Research State is the versioned KG of claims, evidence links, and provenance chains. Formalizing it as a published specification is Contribution 1.

Q [S18]: What is the biggest technical risk? Canonicalization quality. If the entity merging agent creates false merges, the graph loses distinguishing structure. The evaluation's Cohen's Kappa check directly stress-tests this.

Q [S19]: How do you handle terminology mismatches across domains? Cross-domain synthesis is a long-term goal requiring shared ontologies or embedding-based alignment. Near-term, each domain graph uses its own vocabulary; cross-domain linking is deferred to future work.

References

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